Chao Zhang

LLM Inference | Distributed GPU Systems | High-Performance Computing

Stony Brook, NY | (631) 590–9535 | chao.zhang.1@stonybrook.edu | linkedin.com/in/czhang173

Education

Stony Brook University Expected 2028
Ph.D. in Applied Mathematics and Statistics *Stony Brook, NY*

Tongji University 2015
B.S. in Engineering Mechanics *Shanghai, China*

Industry Experience

Machine Learning Engineer Intern — LLM Inference May 2025 – Aug 2025
SiliconFlow *Beijing, China*

- Designed **Snake**, a lightweight speculative decoding architecture for **QwQ-32B and Kimi-K2-Instruct**, spanning draft-model design, training, and inference integration.
- Built a **multi-GPU distributed training framework** from scratch, separating target-model feature generation and draft-model optimization across producer–consumer GPU groups.
- Reduced **time per output token (TPOT) by up to 25%** with no observed time-to-first-token (TTFT) regression in inference benchmarks.

Research Experience

Topology-Aware Distributed GPU Reconstruction 2025 – Present

- Developed a distributed framework for coupled, five-dimensional MRI reconstruction, scaling to **36 nodes / 144 GPUs** and reaching **approximately 1–3-minute solver runtimes**.
- Designed three-level parallelization across motion, echo, and coil dimensions: a **2D inter-node process grid** with **MPI** halo exchanges and node-local coil reductions over **NCCL/NVLink**.
- Overlapped nonblocking communication with GPU computation, achieving a **26% speedup** over synchronous execution at 144 GPUs while preserving numerical accuracy.
- Distributed data and operators across nodes to enable enlarged-field-of-view workloads that exceeded single-node GPU memory.

Variational Regularization and Optimization for Inverse Problems 2026 – Present

- Investigating **Mumford–Shah** variational methods for joint reconstruction of image intensity and edge structures.
- Studying lifted convex relaxations and numerical solution strategies, with total variation regularization as a comparison baseline.

Robust Numerical Solvers for Multi-Species Equilibrium Systems 2022 – 2024

- Developed a piecewise nonlinear solver that combines analytical structure with region-dependent solution strategies to improve stability near singular and ill-conditioned regimes.
- Reduced runtime from **over one hour to under 20 minutes** while preserving the accuracy required by the downstream simulation workflow.

Publication

C. Zhang, X. Jiao, M. Nash, R. Schaake, and Y. Kee. "Large-Scale Distributed GPU-Accelerated Respiratory Motion-Resolved Reconstruction of 3D non-Cartesian mGRE MRI." **MICCAI 2026**.

Patent Application

C. Zhang, Y. Li, and Z. Xie. "Accelerated Inference Method and System for Large-Scale Autoregressive Language Models." Chinese Patent Application **CN202511586683.8**, 2025.

Technical Skills

Languages: Python, C++, C, MATLAB
ML & GPU Computing: PyTorch, TensorFlow, CUDA, CuPy
Distributed Computing & Tools: MPI, NCCL, OpenMP, Linux, Slurm, Git
Methods: Speculative decoding, distributed training, numerical linear algebra, numerical optimization